

60 WATTS LINEAR AMPLIFIER

The 60 Watt linear amplifier is simple all solid state circuit using power mosfet IRF840. The IRF series of power transistors are available in various voltage and power ratings. A single IRF840 can handle maximum power output of 125 watts. Since these transistors are used in inverters and smps they are easily available for around Rs: 20/-.

The IRF linear amplifier can be connected to the out put of popular VWN-QRP to get an output of 60 Watts. The circuit draws 700 ma at 60 Volt Vcc. Good heat sink is a must for the power transistor.

Alignment of the circuit is very easy. Connect a dummy load to the out put of the circuit. You can use some small bulb like 24V 6Watts as the dummy load. I have even used 230V 60Watts bulb as dummy load with my IRF840 power amplifier working at 120Volts. Adjust the 10K preset to get around 100 ma Drain current. I used gate voltage of 0.8V with my linear amplifier. A heigh gate voltage can make the power transistor get destroyed by self oscillation. So gate voltage must be below 2V and fixing at 1V will be safe.

Bifalar transformaer T1 is wound with 8 turns 26SWG on 1.4 x 1 balun core. The coil on the drain of IRF is 3 turns 20 SWG wound on 4 number of T13.9 torroids (two torroids are stacked to form a balun core). The RFC at the Vcc line is 20 Turns 20 SWG wound on T20 torroid.

